Actual local-series bridge for the common rational denominator

Producer proof report; unsigned PCC standing

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The actual common denominator now has an integral formal-series inverse, connected to the imported kernel definition at every scale and source node. The earlier generic polynomial-degree bound is also connected to the actual kernel. These are symbolic proofs; no new experiment or benchmark was run. All results here carry `[pcc:unverified]` standing pending a trusted signature. The original frozen panel, its losses and its PDF remain intact.

Concrete local ring and actual product identity

Set $N = 2^{a+1}$, $L = 2^a$, and $m, s \in \operatorname{Fin} N$. The coefficient ring is the concrete subring

$$A = \{r \in \mathbb{Q} : |r|_2 \le 1\}.$$

For $j \neq m$, $0 < |m - j| < N$ implies $v_2(m - j) < a + 1$: the integer $2^{v_2(m-j)}$ divides $|m - j|$. Thus $c_{m,j} = N/(m - j) \in A$. The actual source normalizer ρ_m and its reciprocal lie in A; `rhoUnit` packages them as a unit. `LocalIntegral.grid_ratio_integral` proves the grid statement.

From the actual punctured-product definition, for *every rational* z,

$$B_s(m - L + Nz) = \rho_m \begin{cases} 1 & s = m, \\ c_{m,s} z & s \neq m \end{cases} \prod_{j \neq m,s} (1 + c_{m,j} z).$$

`LocalCharts.actual_chart_formula` proves this identity; `localBase_eval` identifies the explicit A-polynomial with the imported `baseKernel`. No unit-normalizer or polynomial-representation hypothesis is delegated to the caller. The earlier module only adds an exposed exact product API; its mathematical definition is unchanged.

Actual denominator and integral inverse

Let $P_m = \sum_s \operatorname{localBase}_{m,s}$. Then

$$P_m(z) = D(m - L + Nz), \qquad D(T) = \sum_s B_s(T), \qquad P_m(0) = \rho_m.$$

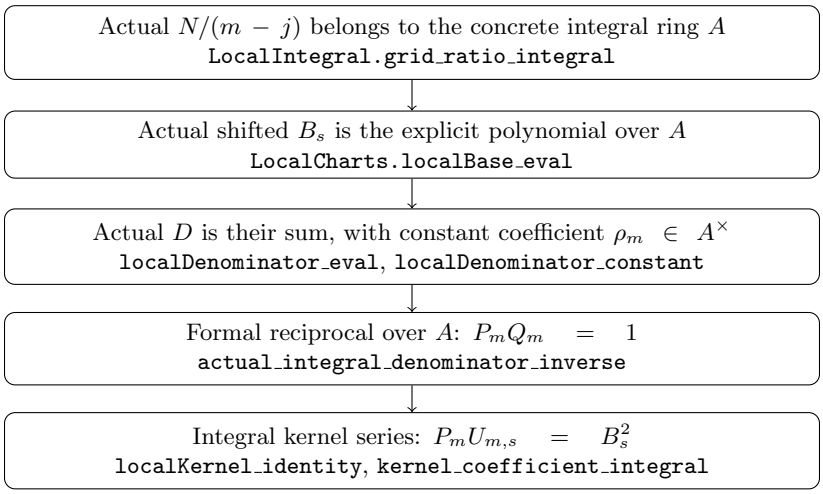
The first equality holds for all rational z. The constant coefficient is a unit, so formal inversion over A constructs $Q_m \in A[[X]]$ with

$$P_m Q_m = Q_m P_m = 1.$$

The theorem `actual_integral_denominator_inverse` states existence of the actual representative and inverse. It assumes neither integrality of an unspecified series nor a denominator-unit hypothesis.

Define $U_{m,s} = \operatorname{localBase}^2_{m,s} Q_m$. `localKernel_identity` proves $P_m U_{m,s} = \operatorname{localBase}^2_{m,s}$. All reciprocal and kernel coefficients belong to A, hence their rational two-adic norms are at most one. This is a coefficientwise formal-series statement; it does not assert analytic convergence at arbitrary rational z. Claims `PCC-20261108-1270b2` and `PCC-20261108-ad84dd` pin these statements and their exact hypotheses.

Source-connected proof chain



The arrows are proved links to actual objects, not measured correlations. The proof map and source are in `lean/LOCAL_SERIES_PROOF_MAP.md`.

Actual global degree and local precision

`GlobalPolynomials.basePolynomial` is the actual puncture polynomial with its actual center denominator. Its evaluation equals imported `baseKernel` for every rational T. Its degree is at most $N-1$, and a finite product of its squares has degree at most $2|E|(N-1)$. This closes the earlier generic-degree bridge; it does not prove fixed-edge homogeneity of the actual recursive contraction program. Claim `PCC-20261108-ee5503` records this scope.

The instantiated inverse coefficient recurrence uses only earlier inverse coefficients. If an ideal I satisfies $E-P_m \in I$ and an approximation has an exact inverse $EF=1$, then

$$F - Q_m = F(P_m - E)Q_m \in I.$$

`LocalPrecision.inverse_jet_stability` takes $I = (2^p, X^q) \subset A[[X]]$. There is no extra precision loss in this algebraic congruence. Both input congruence and $EF=1$ remain explicit hypotheses. Equivalence to a concrete finite buffer, its executable inverse, and full Hermite recovery are still separate. Claim `PCC-20261108-78ae71` pins the conditional theorem.

Checks, provenance and continuation

The pinned Lean project builds these modules; actual source-gate and official import-inclusive fresh-check receipts are recorded in `lean/LOCAL_SERIES_VERIFICATION.md`. Only standard Lean foundations are allowed. All failed builds are retained. KB tools were unavailable (status127); the reviewed primary sources are the pinned Mathlib norm, valuation, polynomial and power-series APIs plus the actual prior kernel and recovery sources. No priority or literature-exhaustion claim is made.

The original eight unsigned ledger events were preserved verbatim and in Git while merging concurrent main registrations. Main remains the byte prefix; unchanged local event bodies were appended by the maintained routine, with old/new hashes in `pcc-reconciliation/reapplied-events.json`.

Next: concrete finite-buffer congruence, actual contraction homogeneity, finite-precision recovery and fully charged construction/child-entry costs with a uniform all-level mode bound. Formal series existence alone supplies neither that evaluator nor its cost recurrence. See `PLAN.md`.